

School Homework Helper

AI Education Platform - Complete Project Requirements

SEO-first platform for Classes 1-12

Document generated from project specification

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1. Project Vision

Build an SEO-first AI education platform for Classes 1-12 - not a single homework page, but a full ecosystem targeting students, parents, teachers, and tutors. Long-term brand vision: 'AI School Hub'.

2. Target Audience

- Students (Class 1-12) - homework help, essays, math, quizzes, science projects
- Parents - homework tips, study plans, class-specific resources
- Teachers / Tutors - lesson plans, report card comments, parent letters (Phase 2)

3. MVP Features (Launch with 6 Tools)

1. Homework Solver: Student enters question; AI explains answer step-by-step.
2. Essay Generator: Generate essays by topic, word count, and class level.
3. Math Solver: Arithmetic, algebra, geometry, and word problems.
4. Quiz Generator: Generate MCQs from any topic.
5. Grammar Checker: Correct English writing.
6. Science Project Ideas: Class-wise project suggestions.

4. Website Structure

Home

AI Tools (6 tools listed above)

Classes (Class 1 through Class 12)

Subjects (Mathematics, Science, English, Hindi, Punjabi, SST)

Blog

About

Contact

5. Technology Stack

Frontend

- Next.js
- Tailwind CSS
- Shadcn UI

Backend

- Next.js API Routes
- Node.js

Database

- PostgreSQL via Supabase

AI

- OpenAI API (primary)
- Gemini API (cheaper backup)

Hosting

- Vercel

6. Database Schema (Minimum)

users

id, name, email, role, created_at

ai_requests

id, user_id, tool_name, prompt, response, created_at

blog_posts

id, title, slug, content, created_at

7. AI Prompt Engineering

Homework Solver

You are a friendly teacher. Explain this homework answer in a way that a Class {grade} student can understand.

Question: {question}

Rules:

- Show step-by-step solution
- Use simple language
- Give final answer
- Add quick revision notes

Essay Generator

Write a {word_count} word essay on: {topic}

For Class {grade} student.

Use: Introduction, Main Body, Conclusion. Easy language only.

Similar tailored prompts are required for Math, Quiz, Grammar, and Science tools.

8. SEO Strategy (Critical for Traffic)

Goal: 500+ SEO landing pages targeting long-tail keywords.

Example Page URLs

- class-5-maths-homework-help
- class-6-science-homework-help
- class-7-english-essay-topics
- class-8-math-word-problems
- class-9-science-projects
- class-10-board-exam-preparation

Target Keywords

- "Class 5 Maths Questions"

- "Class 8 Science Homework"
- "Class 10 Essay Topics"
- "Class 7 English Grammar"

Start with ~100 SEO pages, then scale to 500+.

9. Monetization Strategy

Revenue Sources

- Google AdSense (after launch)
- Affiliate links (books, courses)
- Premium AI credits - Rs. 199/month (after ~20k monthly visitors)
- School advertising (later phase)
- Teacher listings (later phase)

Premium Tier (Post 20,000 Visitors)

Free tier: 5 AI requests per day.

Premium (Rs. 199/month) includes:

- Unlimited AI usage
- Homework PDF Export
- Advanced Math Solver
- Exam Preparation Tools
- Personal Study Planner

10. AdSense Placement Rules

Homepage

- Below hero section
- Between content sections

Blog Pages

- After paragraph 2
- Middle of content
- End of content

AI Tool Pages

- Right sidebar
- Results section
- Footer

Important: Avoid excessive ads at launch.

11. Content Marketing

Publish 5 articles per day. Examples:

- 100 Maths Questions for Class 5
- Best Science Projects for Class 8
- English Essay Topics for Class 6
- Homework Tips for Students
- Board Exam Study Plan

After 6 months: 5 articles/day = 150/month = 1,800/year. This becomes a massive traffic asset alongside SEO pages.

12. Future Expansion (50+ AI Tools)

- AI Worksheet Generator
- AI Report Card Comments Generator
- AI Lesson Plan Generator
- AI Parent Letter Generator

Teacher-focused tools attract dual student + teacher traffic, increasing long-term SEO potential and ad revenue.

13. Growth Targets

Month	Target Visitors
1	500
3	5,000
6	20,000
12	100,000+

14. Development Plan - Where to Start

Week 1 - Foundation (Start Here)

1. Scaffold Next.js app with Tailwind CSS and Shadcn UI
2. Set up Supabase with users and ai_requests tables
3. Build Homework Solver end-to-end (UI, API, OpenAI, DB logging)
4. Design homepage and tool page in Figma
5. Deploy to Vercel for early live URL

Weeks 2-3 - MVP Completion

1. Add remaining 5 AI tools (reuse API + prompt pattern)
2. Build Classes and Subjects hub pages
3. Add About and Contact pages
4. Implement basic authentication (needed before premium)

Week 4 - SEO and Content

1. Generate first 20-50 SEO pages (class + subject combos)
2. Launch blog with CMS or MDX; publish first 5-10 articles
3. Add meta tags, sitemap, and structured data

Month 2+

1. Scale to 100+ SEO pages
2. Apply for Google AdSense
3. Add rate limiting (5 free requests/day)
4. Plan and launch premium tier near 20k visitors

15. Initial Setup Commands

```
npx create-next-app@latest school-homework-helper
```

```
npm install tailwindcss openai @supabase/supabase-js lucide-react zod
```

16. Build Priority Summary

1. Next.js + Tailwind + Shadcn
2. Supabase database schema
3. Homework Solver (one complete tool end-to-end)
4. Deploy to Vercel
5. Remaining 5 AI tools
6. Site structure: Classes, Subjects, Blog
7. SEO pages (batch 1)
8. AdSense + content pipeline
9. Premium features + scale

17. Key Takeaway

The highest-leverage first step is scaffolding the Next.js project and shipping Homework Solver as a complete vertical slice: UI, API, AI integration, database logging, and deployment. Traffic comes from SEO pages and daily blog content; revenue starts with AdSense and affiliates, then premium subscriptions at scale.